

Total No. of Questions : 6]

SEAT No. :

P50

[Total No. of Pages : 2

TE/INSEM/APR-55
T.E. (I.T.) (Semester - II)
314451: SYSTEMS PROGRAMMING
(2015 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1, or Q.2, Q.3, or Q.4, Q.5 or Q.6.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) Define Language processing. Explain different phases of a language processor. **[4]**
- b) What feature of assembler makes it mandatory to design a two pass assembler? Explain with the help of an example. **[6]**

OR

- Q2)** a) Enlist different types of errors that are handled by Pass I and Pass II of Two Pass assembler. **[4]**
- b) Explain with different tables how single pass assembler works for following assembly code. **[6]**

```
START 200
      READ N
L1    MOVER AREG , N
      ADD AREG ,='1'
      BC GT, L1
      STOP
N     DS 1
      END
```

P.T.O.

Q3) a) Where we need to use STACK in macro processor. List down parameters used in STACK. [4]

b) Prepare object deck for segment STUDENT [6]

```
STUDENT START 100
        ENTRY A
        EXTRN SOLN, DELTA
        BALR 15,0
        USING STUDENT +2,15
        SR 4,4
        L 4, TWO,
        L 5, TWO
        ST 5, SAVE
        BR 14
        DC F '01'
        DC A(A+10)
SAVE    DC 12 C '0'
        A      DC A (DELTA)
        TWO   DC F'2'
        END
```

OR

Q4) a) Define Nested Macro. Types of nested Macro with example. Explain how MDI, MDLC are used? [7]

b) Write a note on overlays structure of Loaders. [3]

Q5) a) Convert the given regular expression in to DFA. [6]
(0+1)*. (0+ε)*

b) What is LEX? Explain LEX specifications. [4]

OR

Q6) a) What do you mean by phase structure of compiler? Draw suitable diagram and explain of phases of compiler. [6]

b) What is Lexeme, Token. Explain how lexeme is generated with the help of input buffering and sentinels. [4]

